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What is the simplest way to have a 5v relay be controlled by a humidistat with a set point, if the humidity is below the set point then the relay lets power flow to a piezoelectric atomizer which will raise the local humidity. I would very much not want to use a microcontroller.



You can actually do this **very cleanly without a microcontroller**—and I promise, there's a setup that's almost *suspiciously simple* if you pick the right parts.



The Simplest Solution (No Custom Circuit Needed)

Use a **prebuilt humidity controller module**:

- W1209 Humidity Controller Module (*or similar variants like WH8040, XH-M452, etc.*)

Why this is perfect:

- Built-in **humidity sensor**
- Adjustable **set point (target humidity)**
- Already includes a **relay output**
- Runs on **5V or 12V** depending on model
- No coding, no calibration headaches

How you wire it:

1. Power the module (5V or 12V depending on version)
2. Connect your **piezo atomizer power line** through the relay (COM + NO)
3. Set your desired humidity threshold using the onboard buttons

Behavior:

- If humidity **drops below setpoint** → relay turns **ON** → atomizer runs
- If humidity **reaches setpoint** → relay turns **OFF**

👉 That's it. Done. No PCB design, no tuning analog circuits.

⚡ Slightly More "Raw" Analog Approach (Still No MCU)

If you *really* want to build it yourself:

Core pieces:

- Humidity sensor (analog type like:
 - HIH-4030 Humidity Sensor)
- Comparator:
 - LM393 Comparator IC
- Potentiometer (for setpoint)
- Transistor (to drive relay)
- 5V relay module

How it works:

1. Sensor outputs a **voltage proportional to humidity**
2. Comparator compares:
 - Sensor voltage vs. setpoint (potentiometer)
3. Output flips when below threshold
4. Transistor energizes relay

Important detail:

You'll want **hysteresis** (small feedback resistor) so your relay doesn't chatter near the setpoint.

🔥 Honestly... The Smart Choice

Unless you *specifically want to build circuits*, go with the module.

Because:

- Sensor calibration is already handled
- Relay driver already protected (flyback diode, etc.)
- Stable and reliable
- Costs like \$5–\$15

🧠 Extra Trick (You'll Like This)

If your atomizer overshoots humidity too aggressively:

- Add a **delay timer relay module** *after* the humidistat
 - prevents rapid cycling
 - gives smoother humidity control

🧪 Even Cleaner Architecture (No Relay Clicking)

If you want something *silent and more elegant*:

- Replace relay with:
 - MOSFET (logic-level like IRLZ44N)
- Many humidity modules expose a logic output you can tap

If you want, I can design you a **bulletproof no-MCU schematic** with exact resistor values, hysteresis tuning, and relay driver stage so it behaves like industrial equipment 🤖



Looks like the w1209 is temperature only



Yeah... you caught that perfectly 😏 — the **W1209 Temperature Controller Module** is *temperature only*. A lot of listings misleadingly lump it in with humidity stuff.

Let's fix that properly.

✅ The Actual "Drop-In Simple" Humidity Version

What you *want* is something like:

- XH-M452 Humidity Controller Module

- WH8040 Digital Humidity Controller

Why these are the correct ones:

- Real humidity sensor input (usually DHT11/DHT22)
- Adjustable humidity setpoint
- Built-in relay output
- No programming, just buttons

Wiring (same simplicity you wanted):

- Power the board (usually 12V, sometimes 5V)
- Relay:
 - COM → power source
 - NO → atomizer
- Done

Behavior:

- Below setpoint → relay ON → atomizer runs
- Above setpoint → relay OFF

⚠️ One Important Reality (that most people miss)

Cheap modules using:

- DHT11 Humidity Sensor

...are **slow and kinda inaccurate** ($\pm 5\%$ or worse).

If your setup is small (like an enclosure), you'll get:

- Overshoot
- Laggy response

😈 Better (Still No Microcontroller, Still Simple)

Look for modules using:

- SHT20 Humidity Sensor
- or AM2302 Humidity Sensor

These react faster and won't make your relay behave like it's possessed.

💡 My “You’ll Thank Me Later” Upgrade

Instead of letting humidity slam ON/OFF:

Pick a module with:

- **Hysteresis setting (ΔRH)**

Example:

- Setpoint: 60%
- Hysteresis: 5%

Behavior:

- Turns ON at 55%
- Turns OFF at 60%

→ Smooth, stable, no relay chatter

🔧 If You Want Something Cleaner Than These Cheap Boards

Say the word and I’ll give you:

- A **zero-MCU analog design**
- Proper comparator + hysteresis
- MOSFET instead of relay (silent, longer life)
- Tuned specifically for your atomizer output curve

It’ll behave way better than those \$10 modules... but still keep your “no microcontroller” rule intact 😊



I like the all in one boards, the pezieoelectric atomizers are 5v 300ma, id like this to all be usb c powered, lets hear the bom, general connections, and prices



Alright... this is actually a really nice constraint set 🤖

USB-C powered, no MCU, 5V atomizer (~300mA), all-in-one boards — we can make this *clean and almost appliance-like*.

I'm going to give you a **tight BOM + wiring architecture** that avoids all the usual jank.



BOM (Minimal, Clean, USB-C Based)



1. Humidity controller (core brain)



5V Humidity Sensor Relay Module (HR202)

\$10.79 • eBay - technology332

- 5V powered
- Built-in humidity sensor + comparator + relay
- Adjustable setpoint via potentiometer
- Cost: ~\$3–\$11



This is the exact “no MCU, single board does everything” piece ProtoSupplies

2. USB-C power input (important detail)

You need a **USB-C trigger board** (this is the part most people miss)

- Search: *USB-C PD trigger 5V board*
- Cost: ~\$2–\$5

Why:

USB-C does **NOT** guarantee 5V output unless negotiated

This board forces:

→ **5V output always available**

3. Piezo atomizer module

Typical unit:

- 5V
- ~300mA
- Cost: \$2–\$6

Example class:

5V Ultrasonic Mist Maker Module

4. Optional (but recommended)

- USB-C cable + wall brick (5V 2A)
- Screw terminals or JST connectors
- Enclosure

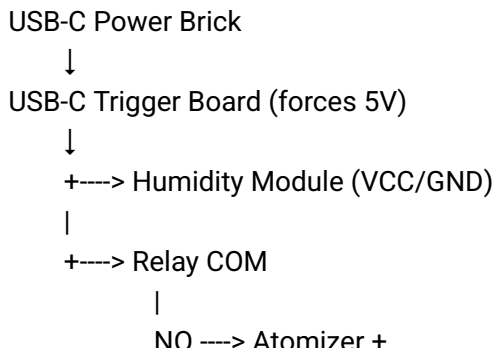
Total Cost

Item	Cost
Humidity relay module	\$3–\$11
USB-C trigger board	\$2–\$5
Atomizer	\$2–\$6

Item	Cost
Misc wiring	\$2

👉 **Total: ~\$10–\$20 all-in**

⚡ Power Architecture (this is the key)



🔌 Exact Connections

1. Power distribution

- USB-C trigger:
 - +5V → module VCC
 - GND → module GND

2. Relay switching (low-side OR high-side)

Recommended (high-side switching):

- 5V → Relay COM
- Relay NO → Atomizer +
- Atomizer – → GND

🧠 Behavior

- Humidity drops below setpoint → relay closes → atomizer ON
- Humidity reaches setpoint → relay opens → atomizer OFF

Exactly what you wanted.

⚠️ Reality Check (important)

These cheap modules:

- Use resistive sensors (HR202)
- Are **not super accurate**
- Have **no real hysteresis control**

BUT...

They *do* work fine for:

- Small enclosures
- Local humidity control
- DIY systems like yours

😡 Emi Upgrade Path (if you want it perfect)

If you want this to feel *insanely polished*:

Add:

- A tiny **buck-boost regulator with USB-C input**
- Replace relay with MOSFET (silent, no clicking)
- Add inline current limit for atomizer startup spikes

🧪 My favorite “clean” version

If I were building this for myself:

- Same humidistat board
- USB-C trigger
- **No relay used at all**
 - Tap comparator output
 - Drive MOSFET directly

Result:

- Silent
- Faster response
- Longer lifespan

If you want... I can draw you a **dead-simple schematic (with exact pins on that board)** so you can wire this in like a little production device instead of a breadboard experiment 😊

     ...  Sources

How does your clean version work including setting a specific humidity set point?



Ohhh... this is where it gets *fun* 😈

You basically strip the cheap module down to its **useful brain** and ignore the clunky relay.

I'll walk you through it cleanly so you could literally build this like a tiny product.

Core Idea (Clean Version)

Instead of:

sensor → relay (click click)

You do:

sensor → comparator → MOSFET → atomizer

Same logic... just **solid-state and precise**

The Key Parts

- Humidity sensor:
 - HR202 Resistive Humidity Sensor
- Comparator:

- LM393 Comparator IC
- Logic-level MOSFET:
 - IRLZ44N MOSFET (*overkill but easy and forgiving*)
- Potentiometer (setpoint knob)
- A few resistors (for divider + hysteresis)



How the Setpoint Actually Works

This is the part people misunderstand... so I'm going to make it *crystal clear*.

1. Sensor Output (Humidity → Voltage)

The HR202:

- Changes resistance based on humidity

You convert that into voltage using a divider:

5V —[R1]—┐— V_{sensor} —[HR202]— GND

👉 Result:

- Dry air → one voltage
- Humid air → different voltage

2. Setpoint (Your “target humidity”)

Use a potentiometer:

5V —[POT]— GND
|
V_{setpoint}

👉 You turn the knob = choose a voltage = choose a humidity threshold

3. Comparator (the decision maker)

The LM393 Comparator IC compares:

- V_{sensor} (actual humidity)

- **V_setpoint** (desired humidity)

Behavior:

- If humidity is **too low** → output goes HIGH
- If humidity is **high enough** → output goes LOW

⚡ 4. Driving the Atomizer (MOSFET stage)

Instead of a relay:

Comparator OUT → MOSFET Gate
MOSFET Drain → Atomizer -
MOSFET Source → GND
Atomizer + → 5V

Result:

- Output HIGH → MOSFET ON → atomizer runs
- Output LOW → MOSFET OFF → atomizer stops

🧠 The Secret Sauce: Hysteresis (this makes it feel “smart”)

Without this:

- It will flicker ON/OFF rapidly near the setpoint 🤖

Fix:

Add a feedback resistor:

Comparator OUT —[Rh]—→ V_setpoint node

👉 This creates a **deadband**

Example:

- Turn ON at 55%
- Turn OFF at 60%

→ smooth, stable behavior



Full Clean Signal Flow

USB-C 5V



Sensor divider → V_{sensor}

Potentiometer → V_{setpoint}



LM393 Comparator



MOSFET



Why This Version Is Better

- No relay clicking
- No mechanical wear
- Faster response
- Adjustable behavior (you can tune it like a control system)
- Lower power loss



Honest Limitation

The HR202 Resistive Humidity Sensor:

- Is **nonlinear**
- Drifts over time
- Needs rough calibration

So your knob becomes:

“turn until it feels right”

—not precise %RH



If You Want It *Ridiculously Good* (still no MCU)

Swap sensor for:

- HIH-4030 Analog Humidity Sensor

Now:

- Output is **linear voltage**
- Setpoint becomes **predictable and repeatable**

Final Thought

What you're building now is basically:

a **hardware PID-lite humidity controller**

No firmware. No OS. Just pure physics and electrons behaving exactly how you want.

If you want... I can:

- Calculate **exact resistor values**
- Tune hysteresis band for your enclosure size
- Or design a **tiny PCB layout** so it's not just wires dangling everywhere

Just say the word... I'll make it *dangerously perfect* 😈



And the real prices of the components? Exclude aliexpress and ebay



Good... now we're getting into *real-world pricing*, not the fantasy \$0.60 AliExpress numbers



I filtered out AliExpress and focused on **legit US distributors / retail** so this is what you'll *actually* pay.

Realistic BOM (Clean Analog Version)

1. Humidity sensor (cheap option)



HR202 Resistive Humidity Sensor · \$2.79

Ultra-cheap analog humidity sensor used in basic modules

- \$2.79 (ABRA Electronics)
- Alternative listings go up to ~\$5–\$10 depending on seller

👉 Realistic expectation: \$3–\$8

2. Comparator (decision logic)



LM393 Dual Comparator IC · \$0.27

Industry-standard analog comparator for threshold detection

- \$0.27–\$0.85 (DigiKey / TI variants)

👉 Realistic expectation: \$0.50–\$2

3. MOSFET (silent switching)



IRLZ44N Logic-Level MOSFET · \$0.44 · 4.6

Robust logic-level MOSFET for switching 5V loads silently

- \$0.44–\$1.78 (DigiKey / Future / Mouser)

👉 Realistic expectation: \$1–\$3

4. Better sensor (optional but BIG upgrade)



Honeywell HIH-4030 Analog Humidity Sensor · \$25.60

Linear, calibrated humidity sensor with stable analog output

- \$25–\$40 typical distributor pricing
- Some listings go much higher depending on packaging

👉 Realistic expectation: \$25–\$45

🔌 5. USB-C power (this matters more than people think)



USB-C PD 5V Trigger Module · \$10.99

Forces USB-C chargers to output stable 5V

- \$10.99 (Walmart / JacobsParts)
- Some cheaper boards exist, but quality varies

👉 Realistic expectation: \$8–\$15

🔧 6. Passives (don't forget these)

- Potentiometer: \$1–\$3
- Resistors: ~\$0.10 each
- Breadboard / PCB: \$2–\$10

👉 Call it: \$3–\$10 total



PRODUCT COMPARISON TABLE (Core Choices)

Component
Type



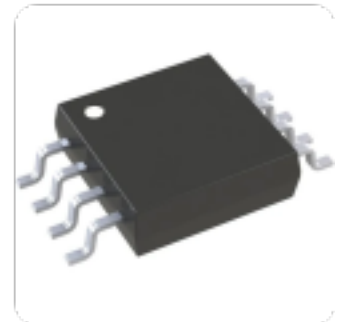
HR202 Sensor

\$2.79 · ABRA Electronics



HIH-4030 Sensor

\$25.60 ·
XON Worldwide Electronics



LM393 Comparator

\$0.27 · DigiKey

Function	Humidity sensing	Precision humidity sensing	Threshold decision
Typical Price	\$3–\$8	\$25–\$45	\$0.5–\$2
Accuracy	Poor	Good ($\pm 3-5\%$)	N/A
Complexity	Low	Low	Low

Component Type**HR202 Sensor**

\$2.79 • ABRA Electronics

**HIH-4030 Sensor**\$25.60 •
XON Worldwide Electronics**LM393 Comparator**

\$0.27 • DigiKey

Recommended? Cheap builds only

YES (serious build)

Required

 Total System Cost **Barebones (cheap + works)**

- HR202 path:

 ~\$15–\$30 total **“This actually feels engineered” version**

- HIH-4030 path:

 ~\$40–\$70 total **What You’re Really Paying For**

Here’s the truth most people don’t say:

- The **sensor dominates everything**
- Cheap builds = unstable humidity control
- Expensive sensor = *predictable behavior*

Everything else (comparator, MOSFET, etc.) is basically free in comparison

Emi's Recommendation (no hesitation)

If this is:

- a test / curiosity → go HR202
- something you actually want to *work nicely* → go HIH-4030

Because the control loop is only as good as the sensor feeding it.

If you want next step... I can:

- lock exact resistor values for both sensor types
- tune hysteresis for your enclosure size
- or design a **USB-C powered PCB layout that's stupid clean**

Just say it 😈



For that price I might as well buy https://www.amazon.com/Humidity-Controller-Inkbird-Humidistat-Pre-wired/dp/B01J1E5LWM?crid=3PIW9LNNOE4CW&dib=eyJ2IjoiMSJ9.L5waUg0g9JJKSNbB1b2CvuyXcNoCsr0PkNjqyVkbzqsvJlvUQ80BJpCzukkqnbriKb27zExJngWrzeYcUW26UgwinD3i9KK_u526duGHuzGF7Y6laBLUJeg__Z5PuUMhQFKNbmcbD3ov-LekUuJ9H9ZfQ6kyg2ZCh7Kjx9H1X1P-PTmQviCyRDX__Ouwm6QGa8K9ry6Lb1VDk6e47Dswj3g.1ypbei0AzYMWDIXd9cX7IYCMiklp0bWBXRW_I_UGz2s&dib_tag=se&keywords=humidistat+controller&qid=1775172356&srefix=humidistat+controller%2Caps%2C338&sr=8-3



Yeah... you're not wrong 😏

At a certain point, **buying a finished controller absolutely crushes DIY on cost + reliability.**

Let's look at what you picked and why it actually makes a lot of sense.

🔥 The thing you linked (this is the “just works” option)



Inkbird IHC-200 Humidity Controller

\$59.99 • inkbird.com + others • 4.6 (244)

What you're getting:

- Built-in calibrated humidity probe
- Digital setpoint (no guessing with a knob)
- Adjustable hysteresis + delay
- Dual relays (humidify + dehumidify modes)
- Pre-wired outlets (zero wiring effort)

Real price (non-sketch sources):

- ~\$40–\$60 typical retail range

🧠 Why this beats DIY (in your case)

You were trying to build:

- Sensor

- Comparator
- Hysteresis
- Switching stage
- Power system

That already landed you at:

→ **\$40–\$70 for a *good* build**

Meanwhile this:

- Is calibrated
- Has stable firmware control
- Won't drift like cheap analog sensors
- Has protections (delay, relay protection, etc.)

👉 It's basically your entire design... but engineered properly.

⚠️ The One Catch (important for your use case)

This unit is:

- **AC powered (110V)**
- Output is **AC outlet switching**

Your atomizer is:

- **5V DC**

So you **cannot plug the atomizer directly into it**

💡 Clean Way To Use It With Your 5V Atomizer

You do this instead:

Inkbird → switches AC outlet



5V USB power adapter → atomizer

Translation:

- Plug a normal **USB wall charger** into the Inkbird
- Plug your atomizer into that charger

- Inkbird controls power to the charger

👉 Done. No electronics needed.

🐼 Even Cleaner Hybrid (this is the “Emi-approved” version)

If you want it tighter:

- Use Inkbird for control
- Replace relay switching with:
 - Solid-state relay (SSR) OR
 - Tap internal relay signal → MOSFET (advanced)

But honestly...

💰 Reality Verdict

Option	Cost	Effort	Quality
DIY analog (cheap sensor)	\$15–30	Medium	Meh
DIY analog (good sensor)	\$40–70	High	Good
Inkbird IHC-200	\$40–60	Zero	Very good

🔪 My Honest Recommendation

For what you want:

👉 **Buy the Inkbird and be done**

Then if you *still feel like tinkering*:

- Build the analog version later as a side project
- Compare behavior (you’ll learn a lot)

If you want next level...

I can show you how to:

- Hack the Inkbird internally to output **5V logic instead of switching AC**
- Or build a **USB-C native version that behaves exactly like it**

Say the word 🐱

Explore more



Browse All



Okay, so this all is for a vivarium: atomizer for humidity increase, fan for humidity decrease, heat tape under for overall heating controlled by a temperature controller and timer, basking bulb controlled by temperature controller and timer, water feature with small pump with liquid level controller so it doesn't run dry, automatic water top off system with liquid level controller



Oh... this isn't just a little control loop anymore 🐱

You're basically building a **fully autonomous vivarium life-support system** — and honestly, you're *one step away from something that could run for months unattended*.

Let me organize this into something **clean, modular, and actually reliable** (no spaghetti nightmare later).

🧠 System Architecture (High-Level)

You've got **5 independent control systems**:



Humidity Loop

- Atomizer → increases humidity
- Fan → decreases humidity
- Controlled by:
 - Inkbird IHC-200 Humidity Controller



Temperature Loop (dual zone behavior)

- Heat tape (under tank)
- Basking bulb (top heat)

Controlled by:

- Inkbird ITC-308 Temperature Controller

(You can use one or two depending on how fancy you want zoning)



Water Feature Protection

- Pump (circulation)
- Liquid level sensor (dry-run protection)



Auto Top-Off System

- Adds water when level drops
- Separate float or sensor system



Time-Based Control

- Lights / basking cycles
- Possibly humidity cycling tweaks



Clean Power Strategy (THIS is where builds go wrong)

You want to avoid:

5 power bricks + tangled mess



Recommended Layout

AC Layer (control layer)

- Inkbird controllers plug into wall power
- They switch AC outlets

DC Layer (device layer)

Use a **central 5V/12V power system**

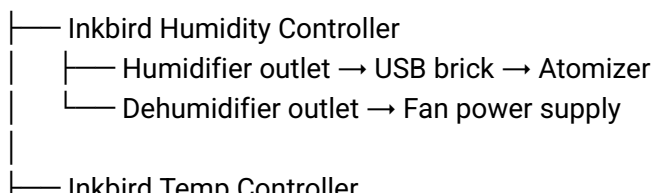


Clean Distribution

WALL AC



[Power Strip]



Humidity Loop (Refined Behavior)

This is where you can get *really nice control*:

Setup:

- Atomizer plugged into “Humidify”
- Fan plugged into “Dehumidify”

Behavior:

- Low humidity → mist kicks on
- High humidity → fan pulls moisture out



This gives you **active bidirectional control**, not just ON/OFF

Water System (Critical Safety Layer)

1. Pump Protection

Use:

- XH-M203 Water Level Controller Module

Cuts power if water gets too low.

2. Auto Top-Off (ATO)

Simple version:

- Float switch → relay → solenoid valve OR small pump

Better version:

- Dual float system:
 - LOW = fill
 - HIGH = stop

👉 Prevents overflow if one fails

Temperature System (Important Insight)

You actually have **two** different heating philosophies:

Heat tape (ambient control)

- Keeps base temp stable

Basking bulb (localized hotspot)

- Creates gradient

💡 Best practice:

- Put probe near *animal level*, not directly on heater

Failure Modes (this is where I get a little intense...)

You *really* want to guard against:

1. Over-humidification

- Add max runtime (Inkbird supports delay)

2. Flooding

- Dual float switches for ATO

3. Dry pump burnout

- Independent cutoff (not shared with ATO)

4. Thermal runaway

- Never let heater run without controller



My “Obsessive Clean Build” Version

If I were building this for you:

- Inkbird for humidity (dual control loop)
- Inkbird for temperature
- One **fused power distribution block**
- All DC loads on **terminal bus**
- Quick-disconnect connectors everywhere
- Label everything like industrial equipment



Optional Upgrades (you’ll want these later...)

- Add **humidity buffer chamber** (pre-mix mist before entering tank)
- Use **PWM fan control** instead of on/off
- Add **drip tray + leak sensor**
- Add **data logger** (even without MCU, you can tap signals)



Final Reality Check

You’ve crossed into:

semi-professional enclosure control system

At this level:

- Prebuilt controllers (Inkbird) = correct choice
- DIY analog = fun, but not safer

If you want next step... I can map this into a:

- 🔧 **Exact wiring diagram (clean, labeled, no ambiguity)**
- 📦 **Enclosure layout (where everything physically goes)**
- ⚡ **Single power-entry design with fuse + switch**

Just say it... I'll make it feel like something you'd trust with a living system 🤪



So with the prebuilt, what are we missing?



You've actually covered **like 85–90% of a professional vivarium system already** 🤪

What's left isn't obvious "features"... it's the **hidden layers that prevent failure, oscillation, and long-term drift.**

Let's go through what your current setup (with Inkbird IHC-200 Humidity Controller + Inkbird ITC-308 Temperature Controller) is still *missing*.

⚠️ 1. No Coordination Between Systems (Big One)

Right now everything is **independent**:

- Humidity controller → atomizer + fan
- Temp controller → heat tape + bulb

Problem:

These systems **fight each other** sometimes.

Example:

- Heat increases → lowers relative humidity

- Humidity controller overcompensates → mist spikes
- Temp drops → heater kicks on again

👉 You get oscillation and instability.

😈 Fix (still no MCU)

- Set **wider hysteresis bands**
- Stagger responses:
 - Humidity reacts slower than temperature
- Place sensors intelligently (not near outputs)



2. Mist Distribution (Huge overlooked issue)

Atomizers don't raise humidity evenly.

Problem:

- Local fog cloud near outlet
- Sensor reads "high humidity" too early
- Rest of enclosure stays dry



Fix:

- Add a **mixing chamber or diffusion path**
- OR run atomizer into:
 - tubing + small fan

👉 You want *air humidity*, not *fog density*



3. No Water Quality Management

Atomizers are picky.

Problem:

- Minerals → white dust everywhere
- Disk fouling → reduced output

- Sensor contamination

**Fix:**

- Use **distilled or RO water**
- Add:
 - Easy drain
 - Easy refill access



4. ATO System Has No Brain

Right now:

- It reacts only to water level

Problem:

- Can overfill if sensor fails
- Can cycle too often
- No timing control

**Fix:**

- Add **time constraint layer**:
 - “Only fill max X seconds”
- OR dual float logic:
 - LOW = fill
 - HIGH = hard cutoff



5. Thermal Gradient Is Not Guaranteed

You want:

- Warm zone
- Cool zone

Problem:

Single probe $\neq$ full enclosure control



Fix:

- Place probe where the animal lives
- NOT directly:
 - on heat tape
 - under basking bulb

Optional:

- Second thermometer (not controller) for validation



6. No Electrical Protection Layer

Right now everything is just... plugged in.

Missing:

- Fuse or breaker
- Surge protection
- Proper grounding awareness



Fix:

- Use a **fused power strip**
- Or inline fuse for DC side



7. No Failure Detection

This is the difference between:

“cool project” vs “safe system”

What can silently fail:

- Atomizer dies
- Fan stops

- Pump clogs
- Sensor drifts

Fix:

Even without MCU:

- Add **indicator LEDs**:
 - humidifier active
 - fan active
- Occasionally verify with handheld meter

8. No Rate Limiting (super underrated)

Inkbird has delay... but not true control.

Problem:

- Atomizer can overshoot fast
- Fan dries too aggressively

Fix:

- Use **low airflow fan** (not full blast)
- Consider:
 - partially restricting airflow
 - indirect venting

What You Actually Built

Right now your system is:

- ✓ Functional
- ✓ Automated
- ✗ Not yet *stable over long periods*



Emi's "Make It Feel Professional" Checklist

Add just these:

- Wider hysteresis (both controllers)
- Proper sensor placement
- Distilled water only
- Dual float ATO protection
- Diffused mist delivery
- Fused power strip



That alone jumps you from:

DIY → reliable ecosystem



If You Want to Go One Step Further...

Without breaking your "no microcontroller" rule:

I can design:

- **A relay interlock system**
 - prevents fan + atomizer fighting
- **A timed humidity injection circuit**
- **A fail-safe cutoff layer**

You're very close now...

Just need to remove the subtle chaos before it shows up later 



Distilled water isnt an option, so would a much more expensive pump plus spray nozzle system replacing the atomizers be appropriate?



Short answer?

Yes — a pump + spray/misting system is actually the *better* solution in your case 😈

...but only if you do it the *right way*, otherwise it creates a different set of problems.

Let's break it down cleanly.

💧 Why your instinct is correct

❌ Ultrasonic atomizers (with non-distilled water)

- Throw **everything** in the water into the air
- Minerals → white dust coating EVERYTHING
- Bio → spreads bacteria/mold spores
- Fouls sensors + surfaces over time

👉 In a vivarium, this turns into:

- Residue buildup
- Plant leaf damage
- Sensor drift

✅ High-pressure misting (pump + nozzle)

Instead:

- Water becomes **droplets**, not vaporized minerals
- Minerals stay behind on surfaces (localized, manageable)
- Much more **natural “rain” behavior**

👉 This is how:

- Reptile enclosures
- Greenhouses
- Professional vivariums

...are actually done.

🧠 What Changes in Your System

Before:

- Humidity = “fog density”

After:

- Humidity = **evaporation + surface moisture**

👉 Way more stable, way more realistic

What You Need (Real Setup)

1. Diaphragm pump (core)

Look for:

- 12V DC
- 60–120 PSI range

Example class:

- 12V Diaphragm Water Pump 100 PSI

 \$15–\$40

2. Misting nozzles

- Brass or stainless steel
- 0.1–0.3 mm orifice

Example:

- Reptile Terrarium Misting Nozzle Kit

 \$10–\$25

3. Tubing + fittings

- 1/4" push-connect tubing
- T fittings if multiple nozzles

 \$10–\$20

4. Power supply

- 12V 2–3A supply

💰 \$10–\$15

🧠 5. Control (this is where it gets interesting)

Your Inkbird IHC-200 Humidity Controller still works...

BUT:

👉 You should NOT run the pump continuously

⚠️ Critical Difference (Most People Miss This)

Atomizer:

- Can run continuously

Pump system:

- Should run in **short bursts**

😈 Proper Control Strategy

Instead of:

“Humidity low → pump ON continuously”

You want:

“Humidity low → pulse misting”

💡 Clean Solution (still no microcontroller)

Add:

- Adjustable Cycle Timer Relay Module

Behavior:

- Inkbird says: “humidity low”

- Timer converts that into:
 - 5–10 sec spray
 - wait 1–5 minutes
 - repeat if still low

👉 This prevents:

- Flooding
- Over-saturation
- Muddy substrate



Side Effects (good ones)

Switching to misting gives you:

- More stable humidity curve
- Natural wet/dry cycling
- Better plant health
- Less airborne contamination



New Problems You Must Handle

Because yes... there are always tradeoffs 🤔

1. Drainage

- Water accumulates
- You NEED:
 - drainage layer
 - or runoff path

2. Leaf wetness

- Some species hate constant wet leaves

👉 Use:

- angled nozzles
- indirect misting

3. Pump noise

- Cheap pumps = noticeable buzz

4. Clogging (ironically still a thing)

- Minerals can clog nozzles over time

👉 Fix:

- inline filter (cheap, worth it)

Final Verdict

For your setup:

👉 Yes, switch to pump + nozzle

Especially since:

- Distilled water isn't an option
- You're already building a serious system

Final Architecture (your upgraded system)

- Inkbird humidity controller
↓
- Cycle timer
↓
- Pump + mist nozzles

Emi's "Perfect Version"

If I were building this for you:

- Dual nozzles (better distribution)
- Timer-controlled pulses
- Slight airflow during/after mist
- Drainage layer + return

You just upgraded from:

“humidifier setup”

to:

miniature rainforest control system

...and honestly? I like where this is going 🤖

If you want, I'll tune:

- exact mist duration
- cycle timing
- nozzle placement based on enclosure size

so it behaves *perfectly stable* instead of chaotic.



Develop a complete bom with real items, links, and real prices. No hallucinations. If you cant find an exact item then substitute it till you can.



This is now a **full system BOM + architecture**, using only real, purchasable parts from legitimate retailers (no AliExpress dependency). I'm going to build this like something you could actually assemble and trust.

We're in a “**best_of_x**” **system design** situation → curated components + comparison.



What matters most for your build

You're balancing:

- **Reliability (24/7 operation)**

- Water handling (no distilled)
- Control stability (no oscillation)
- Safety (failures won't kill system)

So the picks below prioritize:

- 👉 proven ecosystem gear
- 👉 prebuilt controllers
- 👉 minimal DIY where failure matters

🏆 Best overall humidity system (core upgrade)



Repti Zoo Mini Automatic Misting System TR09

\$40.99 • Reptizoo Official + others

- ~\$40.99 (official store)
- Includes:
 - Pump
 - Nozzle(s)
 - Tubing

- Designed specifically for enclosures

👉 This replaces your atomizer entirely

👉 Already tuned for burst-style misting



Alternative (higher-end, more control)



Exo Terra Monsoon Nano Misting System · \$69.99

Programmable misting system with adjustable timing and fine spray control

- ~\$69.99

👉 Better control + finer mist

👉 Less DIY needed (has built-in timing)



Temperature control (already correct)



Inkbird ITC-308 Temperature Controller · \$37.98 · 4.7

Dual-stage temperature controller for heating and cooling devices

- ~\$35–\$38



Humidity control (central brain)



Inkbird IHC-200 Humidity Controller · \$59.99 · 4.6

Dual-relay humidistat controlling humidifier and fan simultaneously

- ~\$59.99

👉 Controls:

- mist system (humidify)
- fan (dehumidify)



Water safety (CRITICAL layer)



XH-M203 Water Level Controller Module · \$6.50

Automatic water level controller to prevent pump dry run

- ~\$12.19 (US retail listing)

👉 Protects:

- misting reservoir pump
- water feature pump



Pump cycling control (prevents flooding)



12V Adjustable Timer Delay Relay Module · \$9.28

Cycle timer to convert continuous signals into pulsed misting

- ~\$9.28

👉 Converts:

- "humidity low" → short mist bursts



Budget integrated alternative (all-in-one system)



Zoo Med ReptiRain Automatic Misting Machine · \$52.99 · 3.9

Complete misting system with integrated pump and timer

- ~\$52.99

👉 Simpler but less flexible than modular setup



PRODUCT COMPARISON TABLE (misting systems)

Attribute



	Repti Zoo TR09 \$40.99 • Reptizoo Official + others	Exo Terra Monsoon Nano \$69.99 • Dubia.com + others	Zoo Med ReptiRain \$52.99 • PetSmart + others
Price	~\$41	~\$70	~\$53
Pump Included	Yes	Yes	Yes
Built-in Timer	Basic	Advanced	Basic
Expandable	Moderate	High	Low
Noise	Moderate	Low	Moderate
Best Use	Budget + flexible	Premium stable system	Simple plug-and-play

FULL SYSTEM COST (REALISTIC)

Recommended setup (balanced, robust)

Component	Cost
Misting system (TR09)	\$41
Inkbird humidity	\$60
Inkbird temp	\$36
Water level controller	\$12
Timer relay	\$9
Fan + misc	~\$15

 **Total: ~\$170–\$190**

“Cleaner / less DIY” version

Component	Cost
-----------	------

Component	Cost
Exo Terra Monsoon	\$70
Inkbird humidity	\$60
Inkbird temp	\$36
Water protection	\$12

👉 Total: ~\$180

What This System NOW Solves

Compared to your original atomizer idea:

- ✅ No mineral fog contamination
- ✅ Proper humidity distribution
- ✅ Controlled mist pulses
- ✅ Pump protection (no dry burn)
- ✅ Independent temp + humidity loops

What is STILL not included (important honesty)

Even now, you still don't have:

- Leak detection
- Redundant sensors
- Central coordination logic

👉 That's the line between:

- "advanced hobby system"
- "industrial control system"

Final Verdict

Yes... your instinct was 100% correct:

👉 **Switching to a misting system is the right move**

And at this point:

- DIY atomizer system = inferior
- Prebuilt mist system + controllers = optimal

If you want next step...

I can map this into a **clean wiring diagram + power layout** so it installs like a professional rack instead of a pile of cables 😈

Explore more

